Applied Physics Letters ARTICLE pubs.aip.org/aip/apl

Identification of charge-state-dependent single-electron levels of color centers at SiO_2/SiC interface by photo-assisted capacitance–voltage measurements

Cite as: Appl. Phys. Lett. **129**, 161601 (2026); doi: 10.1063/5.0351571
Submitted: 8 July 2026 · Accepted: 5 October 2026 ·
Published Online: 19 October 2026

View Online Export Citation CrossMark

Kentaro Onishi,[1,a] Katsuhiro Kutsuki,[2] Masahiro Hara,[1] Heiji Watanabe,[1] and Takuma Kobayashi[1,a]

AFFILIATIONS

[1]Graduate School of Engineering, The University of Osaka, 2-1 Yamadaoka, Suita, Osaka 565-0871, Japan
[2]Toyota Central R&D Labs., Inc., Nagakute, Aichi 480-1192, Japan

[a]**Authors to whom correspondence should be addressed:** onishi@ade.prec.eng.osaka-u.ac.jp and kobayashi@prec.eng.osaka-u.ac.jp

ABSTRACT

Color centers at the SiO_2/SiC interface have attracted attention as single-photon emitters with high photon count rates. However, the single-electron levels involved in the emission process remain unclear. In this study, defect charge states were controlled by applying a voltage and optical excitation during photo-assisted capacitance–voltage measurements. The corresponding single-electron levels were then experimentally evaluated for different charge states. The measurements revealed single-electron levels located at 0.96 and 1.19 eV above the valence-band edge for the singly negatively charged state, and at 0.96 eV for the neutral charge state. These findings provide insights into the electronic structure and emission mechanism of color centers at the SiO_2/SiC interface.

Published under an exclusive license by AIP Publishing. https://doi.org/10.1063/5.0351571

25 October 2026 07:05:52

Single-photon emitters (SPEs) are fundamental elements for quantum technologies such as quantum communication, cryptography, and sensing. SPEs have been realized using various methods, including spontaneous parametric down-conversion,[1,2] single ions and atoms,[3–5] quantum dots,[6,7] and point defects in solids.[8–10] Among these approaches, solid-state color centers are particularly attractive for scalable quantum-device applications because they offer the potential for various fabrication pathways, coupling with photonic structures, and compatibility with electronic devices. Room-temperature operable SPEs based on point defects have been demonstrated in various materials including diamond,[11,12] silicon carbide (SiC),[13–15] and hexagonal boron nitride.[16]

In particular, SiC is promising as a host for color centers due to its well-established crystal growth, conductivity control, and micro-scale processing technologies.[17] Point defects such as silicon vacancies,[13,18,19] divacancies,[14,20] and nitrogen–vacancy centers[15,21] in SiC have been reported to function as SPEs. These bulk color centers have been extensively investigated for applications in quantum sensing and quantum information processing.[10,22] In addition to these bulk color centers, color centers at the SiO_2/SiC interface (interface color centers) have been reported as SPEs.[23] The excited-state lifetimes of interface color centers have been reported to be approximately 2–5 ns,[23] which are even shorter than that of nitrogen–vacancy centers in diamond (∼13 ns at room temperature[24]), indicating their fast emission dynamics and potential for rapid generation of single photons. Interface color centers are readily formed by thermal oxidation of SiC,[23,25] and their density can be controlled by tuning the conditions of oxidation[26,27] and/or post-oxidation annealing,[28–32] with the density evaluated by counting individual bright spots in confocal PL maps. In addition, by incorporating interface color centers into device structures, voltage-dependent modulation of the PL intensity has been demonstrated,[33,34] which offers advantages in practical applications.

However, several properties required for quantum applications remain unclear. To the best of our knowledge, optically detected magnetic resonance (ODMR) has not yet been reported for these color centers, and their responses to magnetic fields and temperature have not been established. In addition, their emission wavelength varies among individual color centers, ranging from approximately 560 to 775 nm,[23,25,26,29–31] indicating that reproducible control of their

optical properties remains a challenge. If their quantum states and responses to external perturbations are clarified and controlled, their near-surface localization may become advantageous for nanoscale sensing, where a short sensor–target distance is important.[35,36] More fundamentally, even the electronic structure of interface color centers, including their defect levels, has not yet been fully established. An understanding of this electronic structure is essential for establishing precise control over their optical properties. In previous studies, correlations between capacitance–voltage (C–V) and PL characteristics have been investigated,[27,30,31] and it was indicated that the charge transition level of the interface color centers lies 0.65–0.92 eV below the conduction-band edge of SiC.[27] However, the charge transition level alone cannot explain optical processes; identifying single-electron levels for each charge state is necessary to understand the optical behavior of interface color centers.

In this study, photo-assisted C–V measurements were performed to experimentally evaluate the single-electron levels associated with interface color centers. The defect charge states were controlled by combining voltage application and optical excitation methods, thereby enabling separate evaluations of the properties of singly negatively charged and neutral charge states. Correlations between the interface state density (D_{it}) and PL intensities were analyzed to identify the single-electron levels of the interface color centers. The PL process and possible origin of interface color centers were investigated on the basis of the experimentally determined energy levels.

4H-SiC(0001) substrates with n-type epitaxial layers (donor density: 1×10^{16} cm^{-3}) were used in this study. Following sequential wet cleaning of the sample surface with acetone, HF, piranha solution, RCA cleaning, and a final HF treatment, SiO_2 films with thicknesses in the range of 9–23 nm were formed by thermal oxidation. The oxidation temperature and oxygen partial pressure [$P(O_2)$] were varied in the ranges of 1400–1600 °C and 0.05%–100%, respectively, resulting in a total of 19 samples with a wide range of defect densities. Confocal PL measurements were performed. A 532-nm excitation laser was focused through an objective lens (NA: 0.9) to a ∼360 nm spot, and photons with wavelengths longer than 600 nm were detected using single-photon counting modules. Metal–oxide–semiconductor (MOS) capacitors with circular aluminum (Al) gate electrodes (diameter: 100 μm) and back-side Al contacts were also prepared for electrical characterization, as shown in Fig. 1(a). Photo-assisted C–V measurements were carried out using a xenon lamp (MAX-303, Asahi Spectra). Monochromatic light with wavelengths of 720, 660, 600, 580, 560, 540, 520, and 500 nm was irradiated through band-pass filters with linewidths of approximately 10 nm at an optical power of about 22 mW. The wavelengths of light were selected to avoid band-edge or free-carrier absorption by SiC.[37] The C–V measurements were performed at 1 MHz with a voltage sweep rate of about 0.5 V s^{-1}, where the back-side contact resistance was low enough to have a negligible impact on the measured capacitance. The SiO_2/SiC interface beneath the gate electrodes was uniformly illuminated via reflection from the backside Al electrodes, as confirmed by identical areal C–V characteristics for devices with different electrode areas after illumination at a representative wavelength.

Charge-state control of the interface color centers was achieved by applying a gate voltage and using optical excitation. Figure 1 shows charge-state control of the interface color centers using the C–V characteristics of a sample oxidized at 1400 °C under $P(O_2)$ of 100% as an example. As shown in Fig. 1(b), a characteristic plateau was evident in the initial forward C–V sweep, indicating electron trapping into the interface states, which changes the defects' charge state to negative. In the reverse sweep, as well as in the second and subsequent bidirectional sweeps, the plateau was no longer observed, and the C–V characteristic shifted in the positive direction. This indicates that the defects remain negatively charged during subsequent C–V sweeps in the dark. The flat-band voltage (V_{FB}) was determined to be about 1.5 V by extrapolating the $1/C^2$–V characteristic in the deep depletion region for the initial forward sweep, which was positively shifted from the ideal value of 0.2 V calculated from the difference in the Fermi level of SiC and the work function of Al. Such a positive shift in the initial V_{FB} suggests that a portion of the defects were already negatively charged before the C–V sweep. When the sweep crossed the plateau, electrons were trapped by the remaining neutral defects, making all of them negatively charged. This resulted in the positive shift in V_{FB} of the reverse C–V curve, as well as in the subsequent

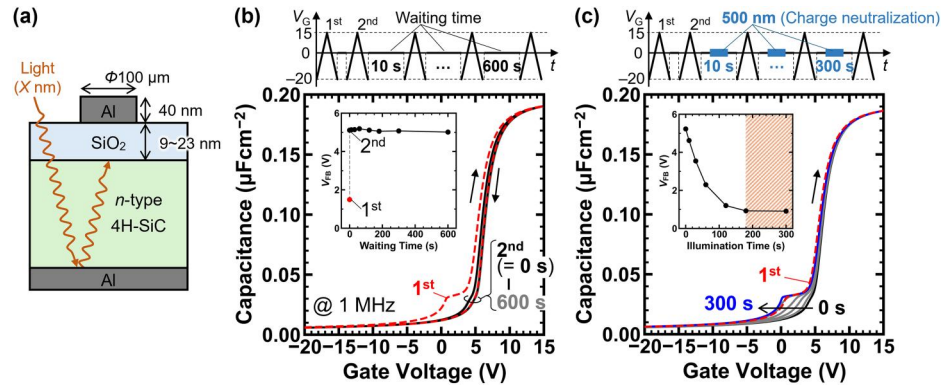


FIG. 1. (a) Schematic cross-sectional view of the sample, showing the device structure and measurement configuration. (b) and (c) 1-MHz C–V characteristics of SiC MOS capacitors fabricated at an oxidation temperature of 1400 °C under an oxygen partial pressure of 100%. Measurement sequences are shown above the figures. (b) Bidirectional C–V characteristics measured in the dark. Inset: the waiting-time dependence of the flat-band voltage (V_{FB}) evaluated from the forward $1/C^2$–V characteristics. (c) Forward C–V characteristics measured after 500-nm light illumination for various time periods at 0 V. The inset shows the illumination-time dependence of V_{FB}. The hatched region indicates saturated V_{FB}.

25 October 2026 07:05:52

Published under an exclusive license by AIP Publishing

measurements ($V_{FB} \sim 5.1$ V). The single, non-multistep plateau in the initial forward sweep implies that the interface defects were singly negatively charged when crossing the plateau. As shown in Fig. 1(c), the neutral charge state of the defects was induced through optical excitation of the trapped electrons by illuminating a 500-nm light at 0 V between the periods of the bidirectional C–V measurements in the dark. Illumination for 180 s or longer shifted V_{FB} toward the ideal value (0.2 V), eventually reaching 0.9 V, indicating that most defects were neutralized after 500-nm illumination for 180 s.

Photo-assisted C–V measurements were performed on the basis of the charge-control technique described above. Figure 2(a) illustrates the measurement principle, and Figs. 2(b) and 2(c) show the results for a sample oxidized at 1400 °C under a $P(O_2)$ of 100% in the singly negatively charged and neutral charge states, respectively. As shown in Fig. 2(a), monochromatic light with different wavelengths was illuminated in order to excite the electrons from the defect levels to the conduction band. This gives us the optical transition energy of the interface defects and allows us to estimate the position of their single-electron energy levels. Illumination was performed under a depletion bias to facilitate drift of the excited electrons from the interface to bulk SiC, preventing their becoming re-trapped by the defects. Here, saturation of the C–V characteristics with respect to illumination time is essential for obtaining well-converged results. We confirmed that an illumination time of 600 s was sufficient for the shortest wavelength used (500 nm) to achieve convergence in the C–V curves of a typical sample. We then examined the results for the negatively charged state, as shown in Fig. 2(b). Light with wavelengths ranging from 720 to 500 nm, corresponding to photon energies of 1.72–2.48 eV, was illuminated at −20 V for 600 s. After the light was turned off, a C–V sweep from depletion to accumulation was performed in the dark. Here, the voltage was not returned to 0 V before the sweep; instead, the sweep was initiated directly from the depletion bias to avoid unwanted electron re-trapping that can occur upon bias release. This procedure was repeated while sequentially decreasing the wavelength from 720 to 500 nm to minimize measurement history effects. To evaluate the neutral charge state as shown in Fig. 2(c), an additional illumination of 500-nm light for 180 s was performed at 0 V prior to the photo-assisted C–V measurements at each different wavelength to initialize the defect charge state to neutral each time. For both singly negatively charged [Fig. 2(b)] and neutrally charged states [Fig. 2(c)], the C–V curves shifted in the negative direction for the deep-depletion region with decreasing light wavelength, or equivalently, increasing photon energy. This is because the electrons were emitted from deeper energy levels to the conduction band as the photon energy increased. The net negative charge decreased as the interface defects lost electrons due to photoexcitation, which caused a negative shift in the C–V curves.

The single-electron levels of the interface defects, i.e., D_{it}, were estimated from the shift in the C–V curves due to photoexcitation. The energy distributions of D_{it} were evaluated from the difference in V_{FB} between C–V curves measured under two adjacent illumination wavelengths (ΔV_{FB}) by using the following expression:[38,39]

$$D_{it}(E_t - E_v) = \frac{C_{ox}\Delta V_{FB}}{eh(\nu_1 - \nu_2)} = \frac{C_{ox}(V_{FB_1} - V_{FB_2})}{eh(\nu_1 - \nu_2)}. \quad (1)$$

Here, C_{ox} is the oxide capacitance per unit area, e is the elementary charge, $h\nu_1$ and $h\nu_2$ are the photon energies of light with two independent wavelengths, and V_{FB1} and V_{FB2} denote the flat-band voltage under each wavelength condition. The energy level of the traps was defined with respect to the valence-band edge, $E_t - E_v$, which was estimated as

$$E_t - E_v = E_g - h(\nu_1 + \nu_2)/2, \quad (2)$$

where E_g is the bandgap of SiC at room temperature (3.26 eV). For the analysis above, we assumed that the electron refilling process from the valence band to the defect levels is negligible. If this process is appreciable, holes will accumulate at the interface under light illumination at the depletion bias, giving rise to a low-frequency capacitance response. The above assumption is supported by the absence of capacitance increase at 1 kHz for a typical sample even under illumination. This was likely due to the small wavefunction overlap between the valence band states and the interface-defect orbitals.

Figure 3 shows the D_{it} distributions obtained for samples oxidized at 1400 °C under different $P(O_2)$ conditions. The D_{it} values increased with $P(O_2)$, which is similar to the behavior of interface color centers where the number of spots increases as the $P(O_2)$ increases.[27] For the singly negatively charged state, two distinct peaks were observed at energies of $E_t - E_v = 0.96$ and 1.19 eV. The similar peak D_{it} values suggest that the defect changes its charge state in one-electron steps when exposed to photon energies

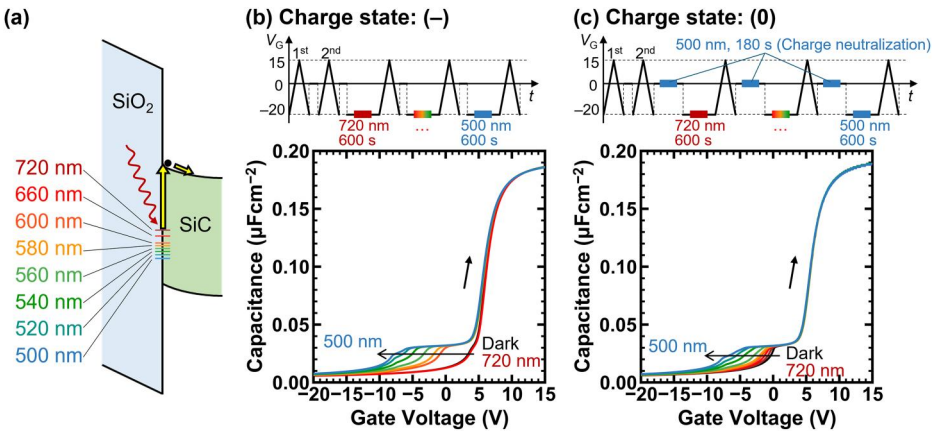


FIG. 2. (a) Schematic band diagram of the SiO_2/SiC interface under illumination with monochromatic light (720–500 nm) at the depletion bias. Photo-assisted forward C–V characteristics for (b) singly negatively charged and (c) neutral charge states measured at a frequency of 1 MHz. Samples were fabricated at an oxidation temperature of 1400 °C under an oxygen partial pressure of 100%. Measurement sequences are shown above of (b) and (c).

25 October 2026 07:05:52

Published under an exclusive license by AIP Publishing

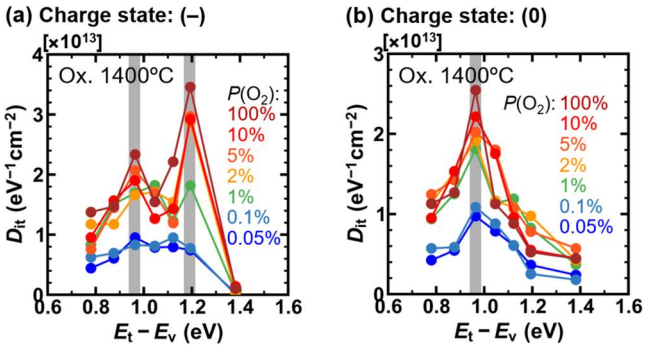


FIG. 3. Energy distributions of interface state density (D_{it}) for SiO_2/SiC structures evaluated by photo-assisted C–V measurements: (a) singly negatively charged and (b) neutral charge states. Trap energy is plotted with reference to the valence-band edge of SiC. Dominant D_{it} peaks are highlighted with gray.

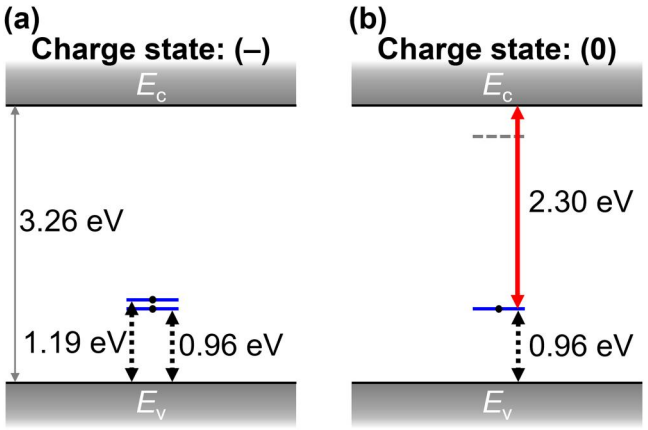


FIG. 5. Schematic single-electron energy-level diagrams of color centers at the SiO_2/SiC interface estimated in this study: (a) singly negatively charged and (b) neutral charge states. In (b), an additional empty level (gray dashed line) is likely located somewhere near the conduction-band edge of SiC, which cannot be identified in this study due to a limitation in the evaluation scheme.

resonant with each defect level. For the neutral state, a single dominant peak was observed at $E_t - E_v = 0.96$ eV. To examine the correlation between D_{it} and the interface color centers, Figs. 4(a)–4(c) plot the peak D_{it} values of each energy level as a function of integrated PL intensity. Here, the integrated PL intensity was evaluated by summing the PL intensities of the pixels in the mapping area (30×30 μm^2). Strong correlations with correlation coefficients (r) higher than 0.8 were obtained between all the D_{it} peaks and the integrated PL intensity. Note that correlation coefficients were much smaller for D_{it} at other out-of-peak energy positions: in the range of 0.31–0.78 for the singly negatively charged state and 0.41–0.74 for the neutral charge state. These results indicate that the detected D_{it} peaks [Figs. 4(a)–4(c)] are those of interface color centers. The slightly lower coefficients, remaining around 0.8 rather than unity, suggest that the energy levels of the interface color centers exhibit a finite distribution, possibly arising from variations in the local environment (e.g., strain) around each defect. Since the present D_{it} measurements probed an ensemble of defects under the gate electrode, the observed peaks should be regarded as statistical representations of the energy levels rather than the exact energy level of a single interface color center. In addition, the oxygen partial pressure appears to have a more significant influence on the formation and properties of the interface color centers than the thermal oxidation temperature, consistent with our previous study.[27]

Based on these results, Fig. 5 describes the energy-level structures of interface color centers: (a) singly negatively charged and (b) neutral charge states. Note that since the spin configuration of the defects could not be identified in this study, the spin majority and minority channels are not distinguished in the figures. In addition, the defect levels are drawn within the bandgap of SiC, as the polarization characteristics of the interface color centers[23,25,26] suggest that the emitters reside in the SiC bulk region adjacent to the interface. The negatively charged state has two distinct energy levels located at $E_t - E_v = 0.96$ and 1.19 eV which are both occupied by electrons, while the neutral charge state has a single occupied level at $E_t - E_v = 0.96$ eV. Since the defect can become negatively charged when the Fermi level is shifted, it is reasonable to assume that, in its neutral charge state, an empty level is located somewhere near the conduction band, as illustrated in Fig. 5(b). However, empty states cannot be directly identified in the photo-assisted C–V measurements, and therefore the exact energy level of such an empty state remains unknown. Since two occupied levels are observed in the singly negatively charged state, it is likely

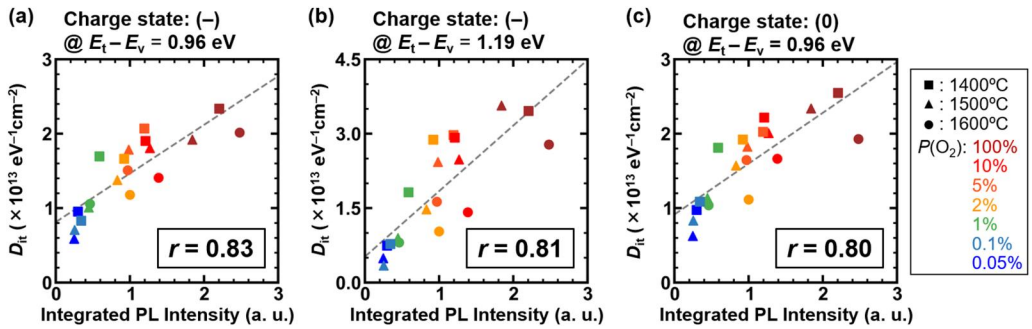


FIG. 4. Correlation between integrated PL intensity and D_{it} peak values located at energy levels of (a) $E_t - E_V = 0.96$ eV and (b) $E_t - E_V = 1.19$ eV in the singly negatively charged state and an energy level of (c) $E_t - E_V = 0.96$ eV in the neutral charge state. Correlation coefficients (r) are indicated in each figure.

25 October 2026 07:05:52

Published under an exclusive license by AIP Publishing

TABLE I. (0/−) charge-transition level of SiO_2/SiC color center identified in our previous work,[27] and occupied single-electron levels for (0) and (−) charge states identified in the present study, together with those of candidate defects in bulk SiC from earlier theoretical calculations.[40,42,43] Energy levels are referenced to the valence-band edge (E_v) of SiC. Single-electron levels are limited to the experimentally accessible energy range ($E_t - E_v \geq 0.78$ eV). Majority and minority spin channels are denoted by ↑ and ↓, respectively, while (↑, ↓) denotes a spin-degenerate level. Single-electron levels for $(O_C)_2V_{Si}$ and $O_C(C_{Si})_2$ defects are from unpublished calculations. For V_C, single-electron energy levels are extracted from Fig. 3 of Ref. 42.

Defect	(0/−) charge transition level (eV)	(0) charge state: occupied single-electron levels (eV)	(−) charge state: occupied single-electron levels (eV)
SiO_2/SiC color center (experimental)	2.34–2.61[27]	0.96	0.96, 1.19
$(C_2)_{Si}$ (k)	2.40[40]	1.63 (↑)[40]	...
$(C_2)_{Si}$ (h)	2.52[40]	1.57 (↑)[40]	...
V_C (h)	2.50[42]	1.2 (↑, ↓)[42]	1.6 (↑, ↓), 2.1 (↑)[42]
$(O_C)_2V_{Si}$	3.13[43]	1.46 (↑, ↓)	1.46 (↑, ↓), 3.00 (↑)
$O_C(C_{Si})_2$	2.80[43]	1.14 (↑), 1.70 (↑)	1.50 (↑), 2.29 (↑), 2.34 (↑)

that, upon capturing an electron, the empty level in the neutral state shifts downward to $E_t - E_v = 1.19$ eV. Since the C–V curve is in the depletion condition at 0 V, it appears that the PL from the interface color centers occurs in their neutral charge state. In this charge configuration, the energy difference is 2.30 eV between the conduction-band edge and the single-electron level located at $E_t - E_v = 0.96$ eV. Since the emission energy is generally smaller than the energy difference in the single-electron levels, this does not contradict the experimentally observed PL energy of interface color centers (1.6–2.2 eV[23,25,26,29–31]) As discussed earlier, an empty state is plausibly located near the conduction-band edge of SiC, and it may also play a role in the optical excitation/emission process. The single-electron levels identified in this study are of importance in clarifying the nature of the interface color centers and ensuring better control over them.

In addition to the single-electron levels identified in this study, we previously reported an acceptor-type charge transition level[27] of the interface color centers. Previously, several defect candidates were discussed in relation to theoretical studies on bulk SiC, such as $(C_2)_{Si}$,[40] V_C,[41,42] $(O_C)_2V_{Si}$,[43] and $O_C(C_{Si})_2$.[43] To compare these candidates with the present experimental results, we here discuss the occupied single-electron levels within the experimentally accessible energy range ($E_t - E_v \geq 0.78$ eV). Table I summarizes the (0/−) charge-transition level of the SiO_2/SiC color center obtained in our previous study,[27] the occupied single-electron levels obtained in the present study, and those of candidate defects in bulk SiC reported in earlier theoretical calculations.[40,42,43] Comparison with the experiment shows that none of these defect models fully agrees with the present results. The predicted single-electron levels generally lie closer to midgap than the experimentally identified levels, and several defects show a different number of single-electron levels from those observed. Among proposed bulk SiC defect models, none can simultaneously account for both the experimentally observed charge-transition levels and the single-electron levels. This suggests that the electronic structure of the interface color centers is strongly influenced by the atomic and environmental conditions specific to the SiO_2/SiC interface. Therefore, first-principles calculations that explicitly include the interface in the model will be beneficial for clarifying their microscopic origin.

In summary, photo-assisted C–V measurements were performed to experimentally identify the single-electron levels associated with color centers at the SiO_2/SiC interface. The charge states of the defects were carefully controlled through a combination of voltage application and optical excitation methods, enabling evaluation of both singly negatively charged and neutral charge states. Single-electron levels were identified at 0.96 and 1.19 eV above the valence-band edge for the singly negatively charged state, and at 0.96 eV for the neutral charge state. The experimentally identified single-electron levels obtained in this work provide useful insights for future first-principles studies aimed at identifying the microscopic structures of color centers present at the SiO_2/SiC interface.

This work was partly supported by JSPS KAKENHI Grant Nos. JP25K01263 and JP25K24649, and JST, PRESTO Grant No. JPMJPR22B5. The authors thank Mr. Sosuke Iwamoto from our group for providing the single-particle energy levels of the $(O_C)_2V_{Si}$ and $O_C(C_{Si})_2$ defects shown in Table I.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Kentaro Onishi: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (lead). **Katsuhiro Kutsuki:** Investigation (equal); Methodology (equal); Resources (equal); Validation (equal); Writing – review & editing (supporting). **Masahiro Hara:** Investigation (equal); Methodology (equal); Resources (equal); Supervision (supporting); Validation (equal); Writing – review & editing (supporting). **Heiji Watanabe:** Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (supporting); Resources (equal); Supervision (supporting); Validation (equal); Writing – review & editing (supporting). **Takuma Kobayashi:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (lead); Resources (equal); Supervision (lead); Validation (equal); Writing – review & editing (lead).

25 October 2026 07:05:52

Published under an exclusive license by AIP Publishing

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

[1]D. C. Burnham and D. L. Weinberg, "Observation of simultaneity in parametric production of optical photon pairs," Phys. Rev. Lett. **25**, 84 (1970).
[2]E. Meyer-Scott, C. Silberhorn, and A. Migdall, "Single-photon sources: Approaching the ideal through multiplexing," Rev. Sci. Instrum. **91**, 041101 (2020).
[3]A. Kuhn, M. Hennrich, and G. Rempe, "Deterministic single-photon source for distributed quantum networking," Phys. Rev. Lett. **89**, 067901 (2002).
[4]H. G. Barros, A. Stute, T. E. Northup, C. Russo, P. O. Schmidt, and R. Blatt, "Deterministic single-photon source from a single ion," New J. Phys. **11**, 103004 (2009).
[5]H. J. Kimble, "The quantum internet," Nature **453**, 1023 (2008).
[6]P. Michler, A. Kiraz, C. Becher, W. V. Schoenfeld, P. M. Petroff, L. Zhang, E. Hu, and A. Imamogùlu, "A quantum dot single-photon turnstile device," Science **290**, 2282 (2000).
[7]P. Senellart, G. Solomon, and A. White, "High-performance semiconductor quantum-dot single-photon sources," Nat. Nanotechnol. **12**, 1026 (2017).
[8]C. Kurtsiefer, S. Mayer, P. Zarda, and H. Weinfurter, "Stable solid-state source of single photons," Phys. Rev. Lett. **85**, 290 (2000).
[9]I. Aharonovich, D. Englund, and M. Toth, "Solid-state single-photon emitters," Nat. Photonics **10**, 631 (2016).
[10]D. D. Awschalom, R. Hanson, J. Wrachtrup, and B. B. Zhou, "Quantum technologies with optically interfaced solid-state spins," Nat. Photonics **12**, 516 (2018).
[11]A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. V. Borczyskowski, "Scanning confocal optical microscopy and magnetic resonance on single defect centers," Science **276**, 2012 (1997).
[12]G. Balasubramanian, P. Neumann, D. Twitchen, M. Markham, R. Kolesov, N. Mizuochi, J. Isoya, J. Achard, J. Beck, J. Tissler, V. Jacques, P. R. Hemmer, F. Jelezko, and J. Wrachtrup, "Ultralong spin coherence time in isotopically engineered diamond," Nat. Mater. **8**, 383 (2009).
[13]M. Widmann, S. Y. Lee, T. Rendler, N. T. Son, H. Fedder, S. Paik, L. P. Yang, N. Zhao, S. Yang, I. Booker, A. Denisenko, M. Jamali, S. A. Momenzadeh, I. Gerhardt, T. Ohshima, A. Gali, E. Janzén, and J. Wrachtrup, "Coherent control of single spins in silicon carbide at room temperature," Nat. Mater. **14**, 164 (2015).
[14]W. F. Koehl, B. B. Buckley, F. J. Heremans, G. Calusine, and D. D. Awschalom, "Room temperature coherent control of defect spin qubits in silicon carbide," Nature **479**, 84 (2011).
[15]J.-F. Wang, F.-F. Yan, Q. Li, Z.-H. Liu, H. Liu, G.-P. Guo, L.-P. Guo, X. Zhou, J.-M. Cui, J. Wang, Z.-Q. Zhou, X.-Y. Xu, J.-S. Xu, C.-F. Li, and G.-C. Guo, "Coherent control of nitrogen-vacancy center spins in silicon carbide at room temperature," Phys. Rev. Lett. **124**, 223601 (2020).
[16]T. T. Tran, K. Bray, M. J. Ford, M. Toth, and I. Aharonovich, "Quantum emission from hexagonal boron nitride monolayers," Nat. Nanotechnol. **11**, 37 (2016).
[17]T. Kimoto and H. Watanabe, "Defect engineering in SiC technology for high-voltage power devices," Appl. Phys. Express **13**, 120101 (2020).
[18]V. Ivády, J. Davidsson, N. T. Son, T. Ohshima, I. A. Abrikosov, and A. Gali, "Identification of Si-vacancy related room-temperature qubits in 4H silicon carbide," Phys. Rev. B **96**, 161114 (2017).
[19]S. G. Carter, O. O. Soykal, P. Dev, S. E. Economou, and E. R. Glaser, "Spin coherence and echo modulation of the silicon vacancy in 4H-SiC at room temperature," Phys. Rev. B **92**, 161202 (2015).
[20]D. J. Christle, A. L. Falk, P. Andrich, P. V. Klimov, J. U. Hassan, N. T. Son, E. Janzén, T. Ohshima, and D. D. Awschalom, "Isolated electron spins in silicon carbide with millisecond coherence times," Nat. Mater. **14**, 160 (2015).
[21]S. I. Sato, T. Narahara, Y. Abe, Y. Hijikata, T. Umeda, and T. Ohshima, "Formation of nitrogen-vacancy centers in 4H-SiC and their near infrared photoluminescence properties," J. Appl. Phys. **126**, 083105 (2019).
[22]N. T. Son, C. P. Anderson, A. Bourassa, K. C. Miao, C. Babin, M. Widmann, M. Niethammer, J. Ul Hassan, N. Morioka, I. G. Ivanov, F. Kaiser, J. Wrachtrup, and D. D. Awschalom, "Developing silicon carbide for quantum spintronics," Appl. Phys. Lett. **116**, 190501 (2020).
[23]A. Lohrmann, N. Iwamoto, Z. Bodrog, S. Castelletto, T. Ohshima, T. J. Karle, A. Gali, S. Prawer, J. C. McCallum, and B. C. Johnson, "Single-photon emitting diode in silicon carbide," Nat. Commun. **6**, 7783 (2015).
[24]I. P. Radko, M. Boll, N. M. Israelsen, N. Raatz, J. Meijer, F. Jelezko, U. L. Andersen, and A. Huck, "Determining the internal quantum efficiency of shallow-implanted nitrogen-vacancy defects in bulk diamond," Opt. Express **24**, 27715 (2016).
[25]A. Lohrmann, S. Castelletto, J. R. Klein, T. Ohshima, M. Bosi, M. Negri, D. W. M. Lau, B. C. Gibson, S. Prawer, J. C. McCallum, and B. C. Johnson, "Activation and control of visible single defects in 4H-, 6H-, and 3C-SiC by oxidation," Appl. Phys. Lett. **108**, 021107 (2016).
[26]M. Kaneko, H. Takashima, K. Shimazaki, S. Takeuchi, and T. Kimoto, "Impact of the oxidation temperature on the density of single-photon sources formed at SiO_2/SiC interface," APL Mater. **11**, 091121 (2023).
[27]K. Onishi, T. Nakanuma, H. Toyama, K. Tahara, K. Kutsuki, H. Watanabe, and T. Kobayashi, "Insight into the energy level structure and luminescence process of color centers at SiO_2/SiC interfaces," APL Mater. **13**, 021119 (2025).
[28]B. C. Johnson, J. Woerle, D. Haasmann, C. T.-K. Lew, R. A. Parker, H. Knowles, B. Pingault, M. Atature, A. Gali, S. Dimitrijev, M. Camarda, and J. C. McCallum, "Optically active defects at the SiC/SiO_2 Interface," Phys. Rev. Appl. **12**, 044024 (2019).
[29]Y. Hijikata, S. Komori, S. Otojima, Y. Matsushita, and T. Ohshima, "Impact of formation process on the radiation properties of single-photon sources generated on SiC crystal surfaces," Appl. Phys. Lett. **118**, 204005 (2021).
[30]T. Nakanuma, K. Tahara, K. Kutsuki, T. Shimura, H. Watanabe, and T. Kobayashi, "Control on the density and optical properties of color centers at SiO_2/SiC interfaces by oxidation and annealing," Appl. Phys. Lett. **123**, 102102 (2023).
[31]K. Onishi, T. Nakanuma, K. Tahara, K. Kutsuki, T. Shimura, H. Watanabe, and T. Kobayashi, "Generation of single photon emitters at a SiO_2/SiC interface by high-temperature oxidation and reoxidation at lower temperatures," Appl. Phys. Express **17**, 051004 (2024).
[32]T. Nakanuma, H. Toyama, K. Tahara, K. Kutsuki, H. Watanabe, and T. Kobayashi, "Design of annealing conditions for the formation of isolated single-photon emitters at SiO_2/SiC interfaces," J. Appl. Phys. **137**, 205702 (2025).
[33]Y. Abe, T. Umeda, M. Okamoto, R. Kosugi, S. Harada, M. Haruyama, W. Kada, O. Hanaizumi, S. Onoda, and T. Ohshima, "Single photon sources in 4H-SiC metal-oxide-semiconductor field-effect transistors," Appl. Phys. Lett. **112**, 031105 (2018).
[34]S. Sato, T. Honda, T. Makino, Y. Hijikata, S. Y. Lee, and T. Ohshima, "Room temperature electrical control of single photon sources at 4H-SiC surface," ACS Photonics **5**, 3159 (2018).
[35]C. L. Degen, F. Reinhard, and P. Cappellaro, "Quantum sensing," Rev. Mod. Phys. **89**, 035002 (2017).
[36]S. Castelletto, C. T. K. Lew, W. X. Lin, and J. S. Xu, "Quantum systems in silicon carbide for sensing applications," Rep. Prog. Phys. **87**, 014501 (2024).
[37]P. Grivickas, K. Redeckas, K. Gulbinas, A. M. Conway, L. F. Voss, M. Bora, S. Sampayan, M. Vengris, and V. Grivickas, "Intrinsic shape of free carrier absorption spectra in 4H-SiC," J. Appl. Phys. **125**, 225701 (2019).
[38]Y. Irokawa, T. Nabatame, K. Yuge, A. Uedono, A. Ohi, N. Ikeda, and Y. Koide, "Investigation of Al_2O_3/GaN interface properties by sub-bandgap photo-assisted capacitance-voltage technique," AIP Adv. **9**, 085319 (2019).
[39]T. Kobayashi, K. Tomigahara, M. Nozaki, T. Shimura, and H. Watanabe, "Separate evaluation of interface and oxide hole traps in SiO_2/GaN MOS structures with below- and above-gap light excitation," Appl. Phys. Express **17**, 011003 (2024).
[40]P. Li, P. Udvarhelyi, S. Li, B. Huang, and A. Gali, "Carbon cluster emitters in silicon carbide," Phys. Rev. B **108**, 085201 (2023).
[41]T. Kobayashi, K. Harada, Y. Kumagai, F. Oba, and Y. Matsushita, "Native point defects and carbon clusters in 4H-SiC: A hybrid functional study," J. Appl. Phys. **125**, 125701 (2019).
[42]J. Coutinho, V. J. B. Torres, K. Demmouche, and S. Öberg, "Theory of the carbon vacancy in 4H -SiC: Crystal field and pseudo-Jahn-Teller effects," Phys. Rev. B **96**, 174105 (2017).
[43]S. Iwamoto, T. Shimura, H. Watanabe, and T. Kobayashi, "Oxygen-related defects in 4H-SiC from first principles," Appl. Phys. Express **17**, 051008 (2024).

25 October 2026 07:05:52

Published under an exclusive license by AIP Publishing